MANISH

Phone: +91 8792084161 | Email: manishgowda2k3@gmail.com | Portfolio Website | LinkedIn: manish-gowda

Technical Skills

Languages : Python, C++, Verilog (HDL).
Developer Tools : VS Code, Linux Ubuntu, Fusion 360, MATLAB, Xilinx Vivado, Ngspice, Magic VLSI, Xschem.

Internship

Evobi Automation Pvt Ltd | Bengaluru *Feb. 2025 – Jun. 2025*

- Worked as a PCB Designer during internship, completing 25+ projects including mini drone, AI bot, and E-Yantra, involving the design and testing of sensor modules such as ultrasonic, IR, and temperature with microcontrollers, verifying functionality under multiple conditions, and creating fully functional 2-layer PCB boards using both SMD and through-hole components.

Airports Authority of India | Mangaluru *Oct. 2023 – Nov. 2023*

- Completed an internship focused on radar systems, gaining knowledge of communication protocols between pilots and air traffic control, understanding equipment upkeep procedures, learning about flight operations, and observing airport systems and equipment including the runway infrastructure.

Education

St. Joseph Engineering College, Mangaluru *Dec. 2021 – May. 2025*
Bachelor of Engineering in Electronics & Communication Engineering *CGPA : 8.81*

Vivekananda Pre - University College, Puttur *Jun. 2019 – Jul. 2021*
12th Grade PCME *91.67%*

Projects

IoT Based Smart Plant Monitoring System *Sep. 2021*

- Designed an automated irrigation system using soil moisture and DHT22 temperature sensors to maintain optimal soil moisture and humidity levels.
- Integrated ESP32 module for remote monitoring and control through a mobile phone interface, enhancing usability and system efficiency.

Vehicle Number Plate Image Processing Using MATLAB *Oct. 2024*

- Implemented a vehicle number plate recognition algorithm using MATLAB for automated number plate detection and extraction.
- Applied a histogram-based number plate detection algorithm using MATLAB, with 75 to 90% accuracy.

Dry Solar Panel Cleaning System *May. 2025*

- Built an efficient waterless solar panel cleaning system to minimize manual effort.
- Developed a mobile app that receives data from a PM2.5 dust sensor for real-time monitoring and remote control; the project received funding of ₹4,500 under the 48th Series of SPP, KSCST.

Differential OTA Design in SkyWater 130nm *Sep. 2026*

- Designed a single-stage PMOS-input differential amplifier in the SkyWater 130 nm process using Xschem and ngspice, achieving 40.4 dB open-loop differential gain at 1.8 V supply.
- Verified all transistors operate in saturation through DC operating-point analysis, with matched input-pair currents and a 1.8 V supply drawing about 31 μA (≈57 μW).

Certification

- Internshala VLSI Design with AI.
- TCS iON Career Edge – Young Professional.
- Google Analytics Certification.
- Kaggle Python Certification.